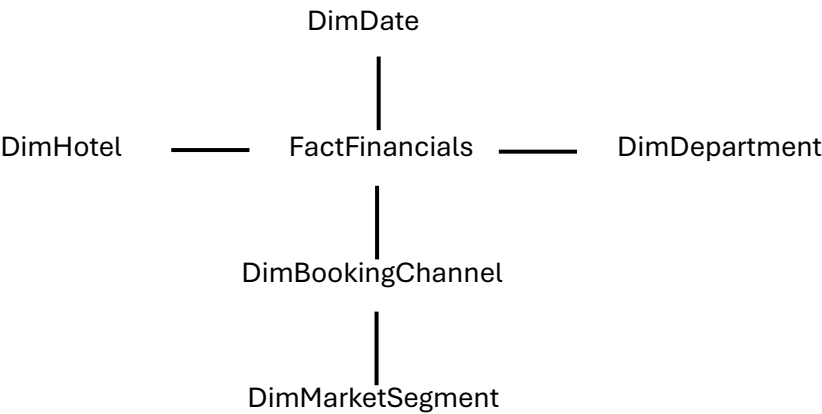


1. Data Model Overview

The solution uses a Star Schema architecture designed for:

- Performance
- Scalability
- Simple reporting
- Efficient DAX calculations

2. Model Structure



3. Fact Table

FactFinancials

Purpose:

Stores all measurable business transactions.

Column	Description
DateKey	Transaction date
HotelID	Hotel reference
DepartmentID	Department reference
ChannelID	Booking source
SegmentID	Customer segment
Revenue	Income generated
Cost	Operating expenses

4. Dimension Tables

DimDate

Purpose:

Provides time analysis.

Includes:

- Date
- Year
- Quarter
- Month
- Week

Used for:

- Trends
 - Growth calculations
 - YTD analysis
-

DimHotel

Purpose:

Enables hotel comparison.

Attributes:

- Hotel name
 - Location
 - Region
 - Hotel type
-

DimDepartment

Purpose:

Provides operational analysis.

Examples:

- Rooms
- Food & Beverage
- Spa
- Laundry
- Parking
- Events

DimBookingChannel

Purpose:

Analyses revenue source.

Examples:

- Direct booking
- Online travel agency
- Corporate
- Travel agent

DimMarketSegment

Purpose:

Customer analysis.

Examples:

- Corporate
- Leisure
- Group
- Business

5. Relationships

From	To	Cardinality
DimDate	FactFinancials	1:M
DimHotel	FactFinancials	1:M
DimDepartment	FactFinancials	1:M
DimBookingChannel	FactFinancials	1:M
DimMarketSegment	FactFinancials	1:M

6. Design Principles

Separation of Facts and Dimensions

Allows:

- Cleaner reporting
- Better performance
- Easier maintenance

Measures Instead of Calculated Columns

Business calculations are created using DAX measures.

Benefits:

- Dynamic filtering
 - Reduced model size
 - Reusable calculations
-

7. Future Enhancements

Potential additions:

- Budget vs Actual table
- Occupancy metrics
- ADR and RevPAR calculations
- Customer lifetime value analysis
- Forecasting model